

# **CS6010 - Social Network Analysis**

## **Unit I - Introduction**

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# **Introduction to Semantic Web**

# Introduction to Semantic Web

- The Semantic Web is the application of advanced knowledge technologies to the Web and distributed systems in general.
- **Information that is missing or hard to access** for our machines can be made accessible using *ontologies*.
- Ontologies are formal, which allows a computer to emulate human ways of reasoning with knowledge.
- Ontologies carry a social commitment toward using a set of concepts and relationships in an agreed way.
- The **SemanticWeb adds another layer on the Web architecture** that requires agreements **to ensure interoperability**.

# Limitations of the current Web

- The current Web has its limitations when it comes to:
  - finding relevant information
  - extracting relevant information
  - combining and reusing information

# What's wrong with the Web?

- The questions below are specific. They represent very general categories of search tasks.
  - **CS6010 social network analysis notes**
  - **Show me photo of paris**
  - **Find new music that I (might) like**
  - **Tell me about music players with a capacity of at least 4GB**

## CS6010 social network analysis notes

- The problem is thus that the keyword *CS6010* is *polysemous*
- The reason is search engines know that users are not likely to look at more than the top ten results.
- Search engines are thus programmed in such a way that the first page shows a diversity of the most relevant links related to the keyword.
- This allows the user to quickly realize the ambiguity of the query and to make it more specific.

## **Show me photo of paris**

- related to the city of Paris and
- showing Paris Hilton

### **Problems:**

- Associating photos with keywords is a much more difficult task than simply looking for keywords in the texts of documents.
- Automatic image recognition is currently a largely unsolved research problem.
- Search engines attempt to understand the meaning of the image solely from its context

- **Find new music that I (might) like**
  - This is a difficult query.
  - From the perspective of automation, music retrieval is just as problematic as image search.
- **Tell me about music players with a capacity of at least 4GB**
  - This is a typical e-commerce query
  - One of the immediate concerns is that translating this query from natural language to the Boolean language of search engines is (almost) impossible.
  - The search engine will not know that 4GB is the capacity of the music player.
- **Google Scholar and CiteSeer are the two most well-known**

#### **Examples**

- Search York Sure, Semantic Web researcher

# How to improve current Web?

- Increasing automatic linking among data
- Increasing recall and precision in search
- Increasing automation in data integration
- Increasing automation in the service life cycle

**Adding semantics to data and services is the solution!**

In each of these cases semantic technology would drastically improve the computer's ability to give more appropriate answers.

# **Development of Semantic Web**

# Research, Development and Standardization

- Web content has been first formulated in 1996 by **Tim Berners-Lee**, the original inventor of the Web.
- Semantic Web has been actively promoted by the **World Wide Web Consortium**.
- Natural Language Processing and Information Retrieval have been applied to acquiring knowledge from the World Wide Web.
- Core technology of the Semantic Web
  - Logic-based languages for knowledge representation
  - Reasoning in the field of AI
- Since the exchange of knowledge in standard languages is crucial for the **interoperability** of tools and services on the Semantic Web, these languages have been standardized by the W3C.

# Technology Adoption

- The Semantic Web was originally conceptualized as an extension of the current Web.
- **Difficulties:** The problem is that as a technology for developers, users of the Web never experiences the Semantic Web directly, which makes it difficult to convey Semantic Web technology to stakeholders.
- The semantic web suffers from **Fax-effect**.
- With the Semantic Web the beginning the **price** of technological investment is **very high**. One has to adapt the **new technology** which requires an investment in learning. **The technology needs time to become more reliable**.
- **Global interoperability** is guaranteed by the standard protocol for communication (**HTTP**).
- In order to exchange meaning : Primitive symbols, Formal Rules
- To follow the popularity of Semantic Web related concepts and Semantic Web standards on the Web, have **executed a set of temporal queries using the search engine Altavista**.

# Five-stage hype cycle of Gartner Research

- technology trigger or breakthrough
  - a frenzy of publicity
  - trough of disillusionment
  - slope of enlightenment
  - plateau of productivity
- Although the word hype has attracted some negative connotations, **hype is unavoidable for the adoption of network technologies** such as the Semantic Web.
  - **The adoption of RDF is lagging behind XML**, even though it provides a better alternative and thus many hoped it would replace XML over time.
  - The vision of Semantic Web as a “web of data”.

# **The Emergence of the Social Web**

# The Emergence of the Social Web

- In 1990s, Web – Combination of phone book and yellow pages. (a mix of individual postings and corporate catalogs)
- This passive attitude toward the Web was broken by a series of *changes in usage patterns and technology* that are now referred to as *Web 2.0*, a buzzword coined by Tim O'Reilly.

# The Emergence of the Social Web

## History of Web 2.0

- Set of **innovations in the architecture and usage patterns** of the Web led to an entirely different role of the online world as a platform for intense communication and social interaction.
- A recent major survey based on interviews with **2200 adults** shows that the internet significantly improves Americans' capacity to **maintain their social networks** despite early fears about the effects of **diminishing real life contact**.
- Successfully activated for dealing with major life situations such as getting support in case of a major illness, looking for jobs, informing about major investments etc.

# The Emergence of the Social Web

## Blogs & wikis

- The first wave of socialization on the Web was due to the appearance of *blogs, wikis and other forms of web-based communication and collaboration.*
- *Blogs and wikis attracted mass popularity from around 2003.*
- For adding content to the Web: *editing blogs and wikis did not require any knowledge of HTML any more.*
- *Blogs and wikis* allowed individuals and groups to claim their *personal space* on the Web and **fill it with content** at relative ease.
- Nowadays the *blogosphere* is widely recognized as a densely interconnected social network through which news, ideas and influences travel rapidly as bloggers reference and reflect on each other's postings.

# The Emergence of the Social Web

## Blogs & wikis

- Wikipedia, the online encyclopedia is outstanding, wikis large and small are used by groups of various sizes as an effective knowledge management tool for keeping records.
- The significance of *instant messaging (ICQ)* is also not just instant communication (phone is instantaneous, and email is almost instantaneous), but the ability to see who is online, a transparency that induces a sense of social responsibility.

# The Emergence of the Social Web

## Social Networks

- The first *online social networks* also referred to as *social networking services*.
- These sites allow users to post a profile with basic information, to invite others to register and to link to the profiles of their friends.
- It leads to discover friends in common, friends thought to be lost or potential new friendships based on shared interests.
- The latest services are thus using user profiles and networks to stimulate different exchanges: **photos are shared in Flickr, bookmarks are exchanged in del.icio.us**, plans and goals unite members at 43Things.
- The idea of **network based exchange** is based on the *sociological observation* that social interaction creates similarity and vice versa, interaction creates similarity: friends are likely to have acquired or develop similar interests.

# The Emergence of the Social Web

## User Profiles

- **Explicit user profiles** make it possible for these systems to introduce **rating mechanism** whereby either the users or their contributions are ranked according to usefulness or trustworthiness.
- **Implementation**
  - Asynchronous JavaScript and XML / *AJAX*
  - Script languages, formats such as JSON, Protocols such as REST
  - RSS Feed
- The results of user experimentation with combinations of technologies are the so-called *mashups*. ***Mashups*** is a websites based on combinations of data and services provided by others.
- *The best example of this development are the mashups based on Google's mapping service such as HousingMaps.*

# The Emergence of the Social Web

## Web 2.0+Semantic Web=Web 3.0?

- **Web 2.0** mostly effects **how users interact with the Web**.
- **Semantic Web** opens new technological opportunities for web developers in **combining data and services from different sources**.
- Web 2.0 is that *users are willing to provide content as well as metadata*.
- The **form articles and facts** organized in **tables** and **categories** in **Wikipedia**, photos organized in **sets** and according to **tags** in **Flickr** or **structured information** embedded into homepages and blog postings using *microformats*.
- Web pages created automatically from a database can **encode metadata in microformats** without the user necessarily being aware of it.
- Semantic technology can help in **matching users with similar interests** as well as **matching users with available content**.

# The Emergence of the Social Web

## Web 2.0+Semantic Web=Web 3.0?

- What Semantic Web can offer to the Web 2.0 community?
  - Standard infrastructure for the building creative combinations of data and services
  - Standard formats for exchanging data and schema information
  - Support for data integration
  - Standard query languages and protocols for querying remote data sources
  - Provide a platform for the development of mashups.

# **Social Network Analysis**

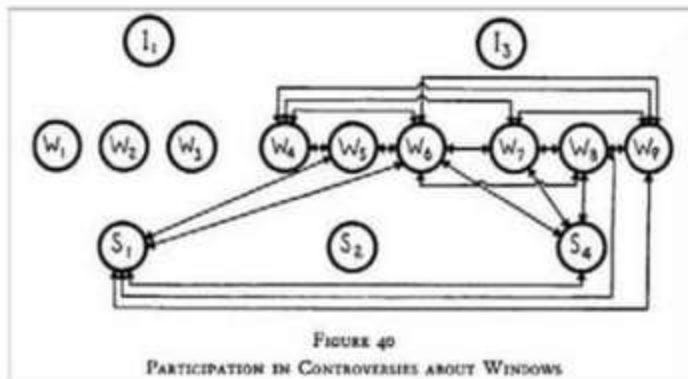
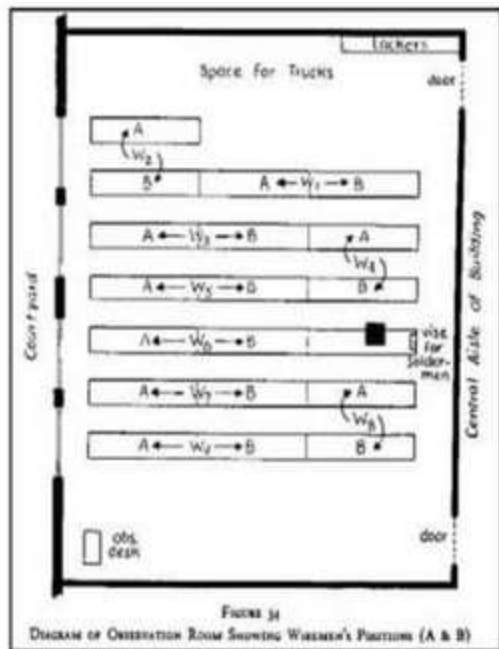
# Social Network Analysis

- Social Network Analysis (SNA) is the study of social relations among a set of actors.
- It focuses on relationships between actors rather than the attributes of individual actors.
- SNA is a different approach to social phenomena and requires a new set of concepts and new methods for data collection and analysis. The concepts and methods of network analysis are grounded in a formal description of networks as graphs.
- *Social role or Social group* be defined on a formal model of networks, allowing to carry out more precise discussions in the literature and to compare results across studies.
- Records of social interaction (publication databases, meeting notes, newspaper articles, documents and databases of different sorts) are used to build a model of social networks.

# Development of Social Network Analysis

- The field of Social Network Analysis today is the result of the convergence of several streams of applied research in sociology, social psychology and anthropology.
- Many **social psychologists** of the 1940s found a formal description of social groups when trying to explain processes of group communication.
- In the mid-1950s **anthropologists** have found network representations useful in generalizing actual field observations
- Researchers from Harvard looked at the **workgroup behavior** (e.g. communication, friendships, helping, controversy) at a specific part of the factory, the bank wiring room.
- The investigators noticed that workers themselves used specific terms to describe who is in “**our group**”. The researchers tried to understand how such terms arise by reproducing in a visual way the group structure of the organization as it emerged from the individual relationships of the factory workers.

# Development of Social Network Analysis



# Development of Social Network Analysis

- In the Southern US researchers looked at the network of overlapping “**cliques**” defined by race and age. They also went further than the Hawthorne study in generating hypotheses about the possible connections between cliques.
- **Example:** They noted that lower-class members of a clique are usually only able to connect to higher-class members of another clique through the higher-class members of their own clique.
- The term “**social network**” has been introduced by Barnes in 1954.
- *Sociogram* was a visual representation of social networks as a set of nodes connected by directed **links**. The **nodes** represented individuals while the edges stood for **personal relations**.

# **Key concepts and measures in network analysis**

# Key concepts and measures in network analysis

- Social Network Analysis has developed a set of concepts and methods specific to the analysis of social networks.
- Network analysis proceed from the global structure of networks toward the measurement of *ego-networks* (*personal networks*), i.e. from the macro level to the micro level of network analysis.

## The global structure of networks

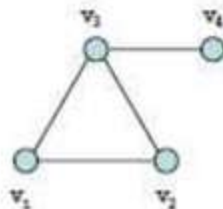
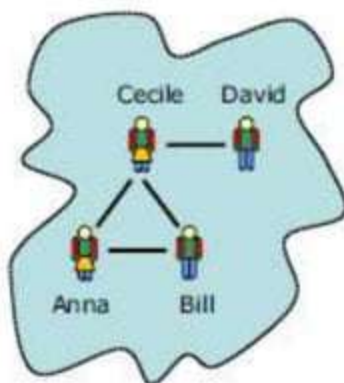
- The social network can be represented as a graph,  $G=(V, E)$  where  $V$  denotes finite set of vertices and  $E$  denotes finite set of edges such that  $E \subseteq V \times V$
- Characteristic Matrix:

$$M := (m_{i,j})_{n \times n} \text{ where } n = |V|, m_{i,j} = \begin{cases} 1 & (v_i, v_j) \in E \\ 0 & \text{otherwise} \end{cases}$$

- For a valued or weighted graph,  $m_{i,j} = w(e) \text{ } (v_i, v_j) \in E, 0 \text{ otherwise}$

# Key concepts and measures in network analysis

## Graph based representation of real world networks



|       | $v_1$ | $v_2$ | $v_3$ | $v_4$ |
|-------|-------|-------|-------|-------|
| $v_1$ | 0     | 1     | 1     | 0     |
| $v_2$ | 1     | 0     | 1     | 0     |
| $v_3$ | 1     | 1     | 0     | 1     |
| $v_4$ | 0     | 0     | 1     | 0     |

# Key concepts and measures in network analysis

- American psychologist Stanley Milgram experiment about the structure of social networks.
- Milgram calculated the average of the length of the chains and concluded that the experiment showed that on average Americans are no more than six steps apart from each other. (i.e) *six degrees of separation*.
- Milgram estimated is the size of the average shortest path of the network, which is also called *characteristic path length*.
- **Geodesic:** Shortest path between two vertices.
- **Diameter:** Longest geodesic in the graph (i.e) Maximum number of steps required between any two nodes.
- **Average Shortest Path:** Average of the length of the geodesic's between all the pair's of vertices in the graph.

# Key concepts and measures in network analysis

- **Clustering**

- It is to measure the degree's of nodes to decide which nodes in a graph tends to be clustered together.
- Clustering for a single vertex can be measured by the actual number of the edges between the neighbors of a vertex divided by the possible number of edges between the neighbors.

- **Clustering coefficient**

Clustering coefficient measure is to quantify how close its neighbor's are to being a complete graph.

$$\text{Clustering coefficient} = 3 * TC / CT$$

TC - Triad Closure

CT - Connected Triple

The clustering coefficient of tree is zero, which is easy to see if we consider that there are no triangles of edges (*triads*) in the graph.

- **Random Graph**

- It can be generated by taking a set of vertices with no edges connecting them.
- Subsequently, the edges are added by picking pairs of nodes with equal probability.

# Key concepts and measures in network analysis

- **Node Degree**

The degree of a node in a graph, is the number of edges incident to the node.

It can be calculated using the formula,

$$\text{Undirected Graph: } N*(N-1)/2$$

$$\text{Directed Graph: } N*(N-1)$$

where N is Number of nodes present in the graph.

- **Node Density**

It is a graph in which the number of edges is close to the maximal number of edges (which is present in the actual graph).

$$\text{Undirected Graph: } (2*E/N)*N-1$$

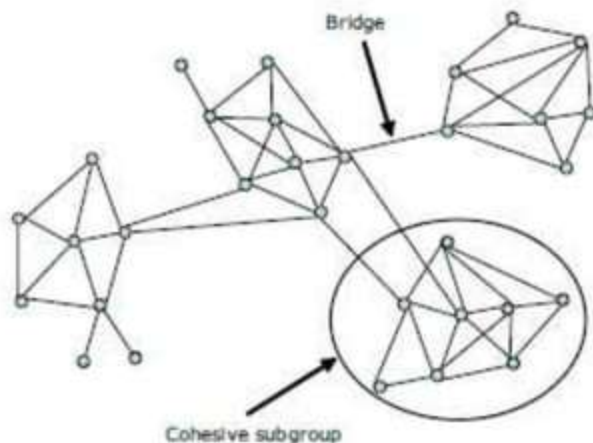
$$\text{Directed Graph: } (E/N)*N-1$$

where E is Number of edges in the network and N is Number of nodes in the network

# Key concepts and measures in network analysis

## The macro-structure of social networks

- To find the global characteristics of social network.
- Most real world networks show a structure where densely connected subgroups are linked together by relatively few bridges.



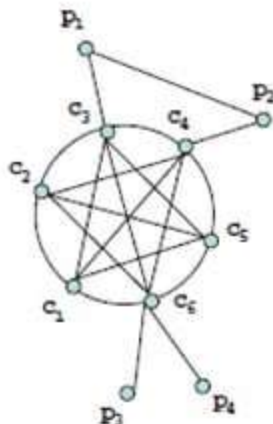
# Key concepts and measures in network analysis

- **Centrality**
  - It is a measure indicating the importance of node in the network.
  - It is further divided into: **Degree Centrality, Betweenness Centrality, and Closeness Centrality**
- **Degree Centrality**
  - It is defined as the number of edges incident upon a node and thus it is usually the first way to calculate the nodes that are most potential to determine other nodes which is present in the network.
  - How many people can reach this person directly?
- **Betweenness Centrality**
  - It is to measure the connectivity of the neighbor's of your node and to give a higher values for nodes which bridge clusters.
  - How likely is this person to be the most direct route between two people in the network?
- **Closeness Centrality**
  - How distinct a node is to the other nodes in the network?
  - How fast can this person reach everyone in the network?

# Key concepts and measures in network analysis

- Core-Periphery (C/P) Structure

- A C/P structure is one where nodes can be divided into two distinct subgroups: nodes in the core are densely connected with each other and the nodes on the periphery, while peripheral nodes are not connected with each other, only nodes in the core.
- The matrix form of a core periphery structure is  $\begin{pmatrix} 1 & \cdot \\ \cdot & 0 \end{pmatrix}$

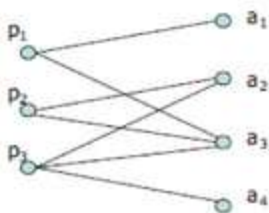


| c <sub>1</sub> | c <sub>2</sub> | c <sub>3</sub> | c <sub>4</sub> | c <sub>5</sub> | c <sub>6</sub> | p <sub>1</sub> | p <sub>2</sub> | p <sub>3</sub> | p <sub>4</sub> |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1              | 1              | 1              | 1              | 1              | 1              | 0              | 0              | 0              | 0              |
| 1              | 1              | 1              | 1              | 1              | 1              | 0              | 0              | 0              | 0              |
| 1              | 1              | 1              | 1              | 1              | 1              | 1              | 0              | 0              | 0              |
| 1              | 1              | 1              | 1              | 1              | 1              | 0              | 1              | 0              | 0              |
| 1              | 1              | 1              | 1              | 1              | 1              | 0              | 0              | 0              | 0              |
| 1              | 1              | 1              | 1              | 1              | 1              | 0              | 0              | 1              | 1              |
| 0              | 0              | 1              | 0              | 0              | 0              | 0              | 1              | 0              | 0              |
| 0              | 0              | 0              | 1              | 0              | 0              | 1              | 0              | 0              | 0              |
| 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 0              | 0              |
| 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 0              | 0              |

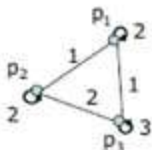
# Key concepts and measures in network analysis

- Affiliation Networks

- Affiliation networks contain information about the relationships between two sets of nodes: a set of subjects and a set of affiliations.
- An affiliation network can be formally represented as a *bipartite graph*, also known as a *two-mode network*.



|       | $p_1$ | $p_2$ | $p_3$ | $a_1$ | $a_2$ | $a_3$ | $a_4$ |
|-------|-------|-------|-------|-------|-------|-------|-------|
| $p_1$ | 0     | 0     | 0     | 1     | 0     | 1     | 0     |
| $p_2$ | 0     | 0     | 0     | 0     | 1     | 1     | 0     |
| $p_3$ | 0     | 0     | 0     | 0     | 1     | 1     | 1     |
| $a_1$ | 1     | 0     | 0     | 0     | 0     | 0     | 0     |
| $a_2$ | 0     | 1     | 1     | 0     | 0     | 0     | 0     |
| $a_3$ | 1     | 1     | 1     | 0     | 0     | 0     | 0     |
| $a_4$ | 0     | 0     | 1     | 0     | 0     | 0     | 0     |



|       | $p_1$ | $p_2$ | $p_3$ |
|-------|-------|-------|-------|
| $p_1$ | 2     | 1     | 1     |
| $p_2$ | 1     | 2     | 2     |
| $p_3$ | 1     | 2     | 3     |

# Key concepts and measures in network analysis

## Personal Networks

- **Structural dimension**
  - Structural dimension of social capital refers to patterns of relationships or positions that provide benefits in terms of accessing large, important parts of network.
- **Relational dimension**
  - Relational dimensions of social capital which concerns the kind of personal relationships that people have developed with each other through a history of interactions.
- **Cognitive dimension**
  - Cognitive dimensions of social capital refers to those resources providing shared representation, interpretations and system of meaning.
- **Structural hole**
  - It occurs in the space that exists between closely clustered communities.

# **Electronic sources for network analysis**

# Electronic sources for network analysis

- **Data collection** using these manual methods are extremely labor intensive and can take up to fifty per cent of the time and resources of a project in network analysis.
- Network researchers are forced to **reanalyze the same data sets** over and over in order to be able to contribute to their field.
- A creative solution to the problem of data collection is to reuse existing **electronic records of social interaction**.
- **Scientific communities** have been studied by relying on publication or project databases showing **collaborations among authors or institutes**.
- **Official databases** on corporate technology agreements allow us to study **networks of innovation**.
- **Newspaper archives** are a source of analysis for social-cognitive networks in **politics**.
- These sources often support dynamic studies through **historical analysis**.
- *e-social science* rely entirely on data collected from electronic networks and online information sources, which allows a complete automation of the data collection process.

# **Electronic discussion networks**

# Electronic Discussion Networks

- The versatility of electronic data is a series of works from the Information Dynamics Labs of Hewlett-Packard.
- **Tyler, Wilkinson and Huberman** analyze communication among employees of their own lab by using the corporate *email archive*.
- They recreate the actual discussion networks in the organization by drawing a tie between two individuals if they had exchanged at least a minimum number of total emails in a given period, filtering out one-way relationships.
- Tyler et al. find the study of the email network useful in identifying leadership roles within the organization and finding *formal* as well as *informal communities*.

# Electronic Discussion Networks

- **Adamic and Adar** revisits one of the oldest problems of network research, namely the question of *local search*: *how do people find short paths in social networks based on only local information about their immediate contacts?*
- Their findings *used to identify* on contacts such as their *physical location and position* in the organization allows employees to conduct their search much more efficiently.
- Discussions are largely in email and to a smaller part on the phone and in face-to-face meetings.
- The studies of **electronic communication networks** based on *email data* are limited by *privacy concerns*.
- **Marc Smith and colleagues** have published a series of papers on the *visualization and analysis of USENET newsgroups*, which predate Web-based discussion forums.

# Electronic Discussion Networks

- **Group communication** and **collective decision taking** in various settings are traditionally studied using much more limited written information such as transcripts and records of attendance and voting.
- The main technical contribution of **Gloor** is a **dynamic visualization of the discussion network** that allows to quickly identify the moments when key discussions take place that activates the entire group and not just a few select members.
- **Gloor** also performs a *comparative study across the various groups* based on the structures that emerge over time.

# **Blogs & Online Communities**

# Blogs & Online Communities

- Content analysis has also been the most commonly used tool in the computer-aided analysis of blogs (web logs).
- **Blogs** are often considered as “*personal publishing*” or a “*digital diary*”.
- Modern blogging tools allow easily commenting and reacting to the comments of other bloggers, resulting in webs of communication among bloggers.
- Blogs make a particularly appealing research target due to the *availability of Structured electronic data* in the form of *RSS (Rich Site Summary) feeds*.
- **RSS feeds** contain *the text of the blog posts* as well as *valuable metadata* such as the *timestamp of posts*, which is the basis of dynamic analysis.
- The 2004 US election campaign represented a turning point in blog research as it has been the first major electoral contest where blogs have been exploited as a method of building networks among individual activists and supporters.

# Blogs & Online Communities

- This fig shows some of the features of blogs that have been used in various studies to establish the networks of bloggers.

Sunday, August 20, 2006

Thinking and berries in time

I has not been blogging much last week, but this is only because I has been writing :) And, the best thing of it is where and how I has been writing.

I'm in Umeå, Sweden, for a workshop, presentation and work/fun with [Stephanie](#). I'm happy I was able to come a few days earlier.



So far it has been almost perfect work-life balance environment. I worked on my own stuff (more productively than in my own office), discussed tons of stuffs with Stephanie (mainly on weblog research, JPL and baking), enjoyed culture and nature, and also that with picking and eating lots of berries.

Some time back [Aldo](#) wrote about thinking locations - places where you can get away from the pressures of the urgent to think your big deep thoughts - I was thinking of it while I enjoyed work and fun here in Umeå.

The social component is very important, and perhaps one of the unique aspect of such a Deep Thought-network: thinkers need on the one hand to be able to concentrate, focus, and withdraw from the world. On the other hand, they very much need to be able to talk with kindred spirits, preferably people working on their own creative projects.

More on <http://thinkingcommunities.wikispaces.com>

Comments | Trackbacks | Link from other address  
View this on the Web

Link to another blog

Link to another blog post

Quote

Links from other blogs

Comments

# Blogs & Online Communities

- **Online community** spaces and social networking services such as *MySpace*, *Live Journal* cater to socialization even more directly than blogs with features such as *social networking (maintaining lists of friends, joining groups), messaging and photo sharing*.
- Most online social networking services (Friendster, Orkut, LinkedIn and their sakes) closely guard their data even from their own users.
- A technological alternative to these centralized services is the FOAF network.
- **FOAF profiles** are stored on the web site of the users and linked together using hyperlinks.
- The drawback of FOAF is that at the moment there is a lack of tools for creating and maintaining profiles as well as useful services for exploiting this network.
- **Advantages**
  - Easy to create and fast
  - Easy to add links, photos, videos
  - It can be used to create community
- **Disadvantage**
  - Generally one author
  - Used for personal opinions and reflection

# **Web based networks**

# Web based networks

- The content of Web pages is the most inexhaustible source of information for social network analysis.
- There are two features of web pages that are considered as the basis of extracting social relations: *links and co-occurrences*.
- The **linking structure** of the Web is considered as proxy for real world relationships as links are chosen by the author of the page and connect to other information sources that are considered authoritative and relevant enough to be mentioned.
- The biggest drawback of this approach is that such *direct links between personal pages are very sparse*: due to the increasing size of the Web searching.



Welcome!

I am a PhD student at the Vrije Universiteit of the Knowledge Representation and Reasoning group of the AI Department and the Communication Sciences department of the Faculty of Social Sciences. My supervisors are Frank van Harmelen (AI) and Jan Klenk (philosophy) (FRO).

I am also co-founder of ZAT, an IT company in consultancy and on-demand software development.

This website is made using a minimal amount of javascript to create one changing button and "clean" output and etc. If you know a way to replace the existing behavior without javascript or if the site looks weird in your browser, please let me know.

This website is best viewed with an expensive monitor and a fast browser.

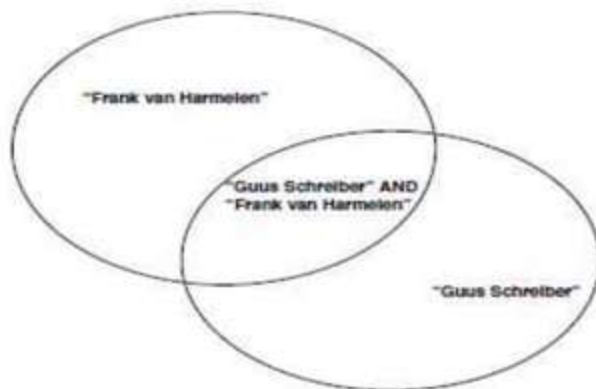
[\[main\]](#)  
[\[research\]](#)  
[\[personal\]](#)  
[\[contact\]](#)

Link to webpage  
of another  
researcher

Co-occurrence  
of names

# Web based networks

- **Co-occurrences of names** in web pages can also be taken as evidence of relationships and are a more frequent phenomenon.
- Extracting relationships based on co-occurrence of the names of individuals or institutions requires web mining.
- **Web mining** is the application of text mining to the content of web pages.
- Using the **search engine Altavista** the system collected page counts for the individual names as well as the number of pages where the names co-occurred.
- **Tie strength** was calculated by dividing the number of co-occurrences with the number of pages returned for the two names individually.



# Web based networks

- **Jaccard-coefficient:** The ratio of the sizes of two sets: the intersection of the sets of pages and their union.
- The resulting value of tie strength is a number between **zero (no co-occurrences)** and **one (only co-occurrences)**. If this number has exceeded a certain fixed threshold it was taken as evidence for the existence of a tie.
- The number of pages that can be found for the given individuals or combination of individuals.
- The reason is that the **Jaccard-coefficient is a relative measure of co-occurrence** and it does not take into account the absolute sizes of the sets.
- A **disadvantage** of the Jaccard-coefficient is that it penalizes ties between an individual whose name often occurs on the Web and less popular individuals.

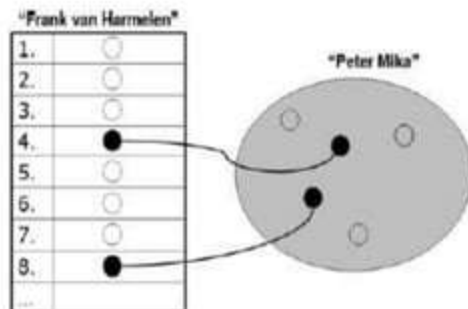


# Web based networks

- There have been several approaches to deal with **name ambiguity**.
- The idea is that **key phrases can be added to the search query** to reduce the set of results to those related to the given target individual.
- When computing the weight of a directed link between two persons. We consider an ordered list of pages for the first person and a set of pages for the second (the relevant set) as shown in Figure.
- There are four different sets: The records which were retrieved, the records which were not retrieved, the relevant records and the irrelevant records.

Recall is defined as:  $\text{Recall} = TP / (TP + FN)$

Precision is defined as:  $\text{Precision} = TP / (TP + FP)$



## Web based networks

- We ask the search engine for the top  $N$  pages for both persons but in the case of the second person the order is irrelevant for the computation.
- Let's define  $rel(n)$  as the relevance at position  $n$ , where  $rel(n)$  is 1 if the document at position  $n$  is the relevant set and zero otherwise ( $1 \leq n \leq N$ ).
- Let  $P(n)$  denote the precision at position  $n$  (also known as  $p@n$ ):

$$P(n) = \frac{\sum_{r=1}^n rel(r)}{n}$$

- Average precision is defined as the average of the precision at all relevant positions:

$$P_{ave} = \frac{\sum_{r=1}^N P(r) * rel(r)}{N}$$

# Web based networks

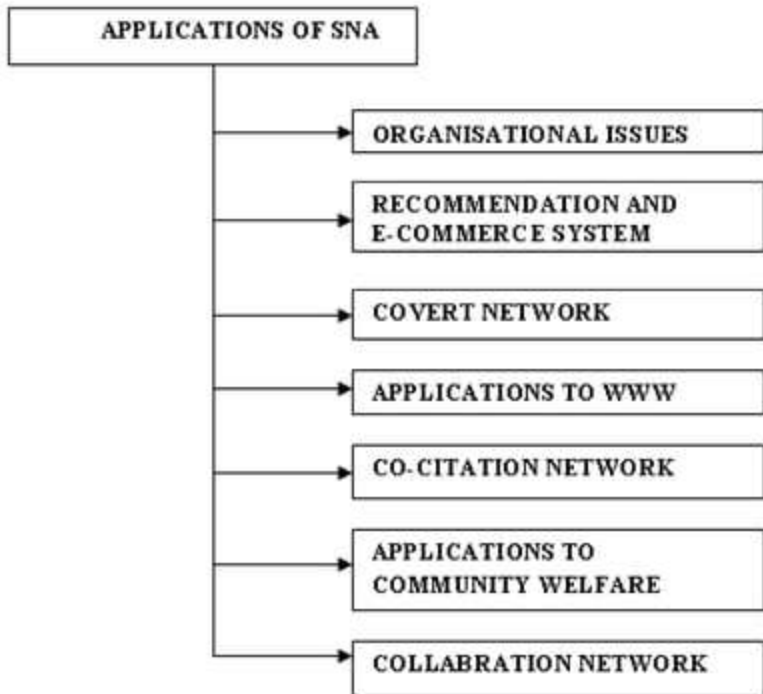
- The **strength** is determined by taking the number of the pages where the name of an interest and the name of a person co-occur divided by the total number of pages about the person.
- *Polysemy* is the association of one word with *two or more distinct meanings*.
- A *polyseme* is a word or phrase with *multiple meanings*.
- The semantic qualities or sense relations that exist between words with closely related meanings is *Synonymy*.

# **Applications of Social Network Analysis**

# Applications of Social Network Analysis

- **Businesses** use SNA to analyze and improve communication flow in their organization, or with their networks of partners and customers.
- **Law enforcement agencies (and the army)** use SNA to identify criminal and terrorist networks from traces of communication that they collect; and then identify key players in these networks.
- **Social Network Sites** like Facebook use basic elements of SNA to identify and recommend potential friends based on friends-of-friends.
- **Civil society organizations** use SNA to uncover conflicts of interest in hidden connections between government bodies, lobbies and businesses.
- **Network operators (telephony, cable, mobile)** use SNA-like methods to optimize the structure and capacity of their networks.

# Applications of Social Network Analysis



# Applications of Social Network Analysis

## Organizational Issues

- Organizational Behavior is the study and application of knowledge about how people, individuals, and groups act within an organization.
- In any organization, cooperation and information sharing among the workers is very important for the success of an organization.
- SNA can also be used to identify the key or central persons of an organization which also helps to understand important to go people in an organization.
  - Team Formation
  - Improved Information sharing
  - Identifying bottlenecks
  - Hidden barriers

# Applications of Social Network Analysis

## Recommendation and E-commerce Systems

- Recommendation systems are web services that provide information about entertainment elements, scientific papers, books, fashion elements such as clothing etc.
- Typically recommendation systems allow users to select and rate items according their own interest and opinion.
- Allowing users to create their own list of items according to their own likes, and allowing user to create their favorites.
- Most of the e-commerce sites such as **Amazon**, **ebay** etc. have their own recommendation systems for recommending customized products to customers and also tries to improve targeted marketing of products.

# Applications of Social Network Analysis

## Recommendation and E-commerce Systems

- Social Network analysis in recommendation systems helps to enhance selling by converting browsers into buyers. Also, these websites acts as recommender agents to learn customers, obtain their preference and provide items of their interest.
- The SNA makes use of various metrics such as centrality, cohesiveness, degree of vertex etc. each may reflect different meaning in recommendation system analysis.
  - Node with high centrality means it has high impact on other nodes.
  - The vertex similarity may be considered as metric to search the individuals having same interest or preference.
  - Cohesiveness property of network defines a group of nodes of network bounded with each other by some relation and may have common characteristics.

# Applications of Social Network Analysis

## Covert Networks

- The covert networks are hidden, the actors of such network does not disclose their information to the external world.
- Covert groups have cellular networks structure which is different from hierarchical organizations. The terrorist and criminal networks are good examples of such networks.
- SNA has been successfully applied to such domains to understand covert cell operations and their organization.
- SNA is applied on terrorism database for predicting node and link, discovering interesting patterns and actors involved in an event.
- Another vital application of SNA for terrorist database is to predict terrorism activities.

# Applications of Social Network Analysis

## Covert Networks

- SNA tools has been used to identify these organization structures and provide critical information for terrorist detection and terrorism prediction.
- Centrality is the most important and widely used measure in SNA used to identify key players in terrorist network.
- To facilitate this, the regular day-to-day activities of the key players are monitored. The hidden actors are discovered by monitoring contact and the extend of contacts of known terrorists with other people.

# Applications of Social Network Analysis

## Web Applications

- Web is being used by different community for various purposes such **academic improvement, knowledge sharing, interest sharing, communication and profiling, research, business** etc. Hence, different techniques are required to improve and optimize the usability of web.
- SNA models web as a graph where web pages are represented as nodes and hyperlinks as edges.
- Node similarity based SNA techniques are employed to classify the web based on its usage and contents in order to understand the scope of domain and density.
- SNA is also used in search engines such as **google** to enhance keyword search quality. Google uses **PageRank** as a measure of popularity, which is obtained by simulating a random walk on network of web pages and computing prestige of web pages.

# Applications of Social Network Analysis

## Community Welfare

- SNA is used to analyze different types of relations such as communication patterns, physical contacts, sexual relationship etc.
- SNA may reveal the patterns of human contact which may lead to spread of disease such as HIV in population.
- SNA to examine and observe farm animal network to identify patterns of disease spread from one animal to another.
- Mass surveillance is done with the purpose of protecting people from criminals, terrorists or political subversives to maintain social control.
- Social Networks which are made for strengthening community resilience against disasters (natural or human-made) can reveal vulnerabilities within a network.

# Applications of Social Network Analysis

## Collaboration Networks

- Collaboration network consists of groups of persons working together to perform particular activity and studying human collaboration is an important topic in sociology.
- The widely studied collaboration network by researchers in context of SNs are Co-authorship collaboration network and movie actor collaboration networks.
- The co-authorship network is
  - To study dynamics in patterns of interactions between educational entities or communities.
  - To study and understand the interdisciplinary research which is key factor for innovation.
- Another type of collaboration network is **knowledge collaboration network**. The information about **Open Source Software** needs to be distributed amongst community or users.

# Applications of Social Network Analysis

## Co-Citation Networks

- Co-citation is used as a measure of similarity between two objects.
- Co-citation analysis helps to understand the status and structure of scientific research.
- Basic two approaches of co-citation are
  - **Author co-citation and**
  - **Document co-citation**
- In the field of methodological evaluation, co-citation analysis has been employed to search for invisible colleges.
- This reveals the research network consisting of different institutions linked to each other informally by having indicators to each others documents/papers which can be used to get group of institutes having similar ongoing research.